

State of the Art in Technology: Cybersecurity Lecture Notes

Open the lecture with this

- Hook: Who run a python program on their computer? Can you delete files by running python? What if someone could tell your computer to run a python code without touching your computer even once?
- Remote code execution is a a phenomena where people are able to tell your computer to run code, without physically touching your computer.
- Examples: run-somewhere that encrypts all files of company's computer, and data-breach where someone steals passwords of all users from a website.

Objectives of the course:

- You will how to hack and protect from hackers.
 - You will learn "How does one computer, can tell another computer to run code, and make it do whatever"
 - You will learn "how can someone steal passwords of all users from a website".
 - You will learn how to hack a website.
 - You will learn how to protect from hackers

[Ethics] Why do we learn this?

- When we write software, we may do mistakes that makes it vulnerable.
- We want you to be aware of the possibilities of your code to act in a way you did not expect.

[Learning objective 1: Basic concepts] What is "RCE"?

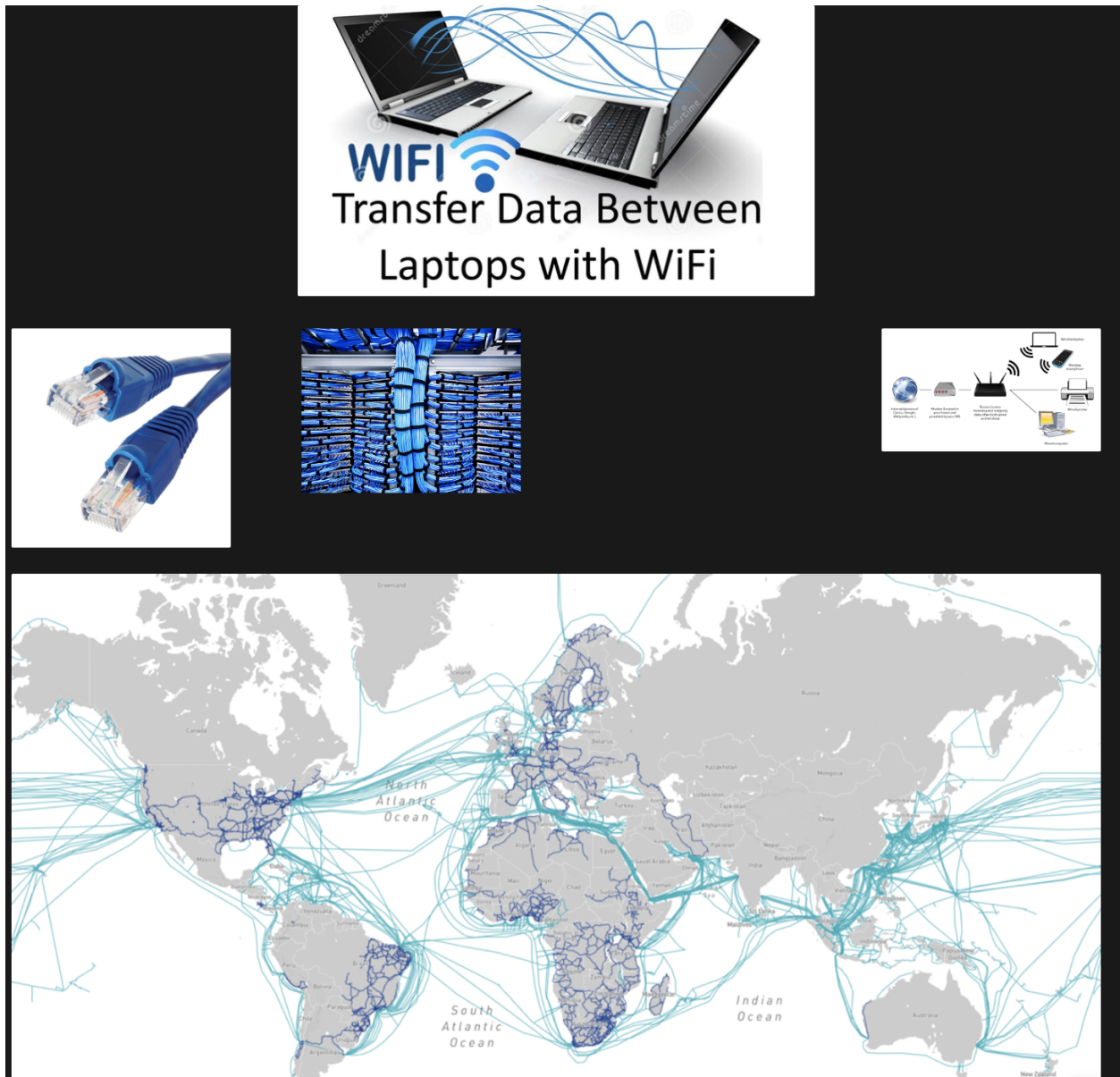
- What is "code execution"?
 - Open "terminal" in your computer. `color 0a; dirtree`
 - You can run "commands/code" on a computer to affect the computer.
- What is "remote code execution"?
 - The ability to run code on a computer, without having physical access to the computer.

- How “remote code execution” is possible?
 - Conceptually, if computer B allows anyone to send it a message, and it takes that **message** and run it, anyone on the internet can execute code on computer B.
- How “remote code execution” actually works?
 - Computers are connected to each other via **network**.
 - Computer A can send a **message** to computer B over the **network**.
 - Computer A can send a **message** to computer B, asking computer B to run a specific command or code.
 - Computer B need to run a software that **listens** to incoming connections from the network.
 - Computer A can send **malicious message** that causes Computer B to run any code Computer A wants.
- Why is it hard to send a **message** to any computer, and ask it to run code for you?
 - If computer A sends a **message** to a software that runs on Computer B, it does not guarantee that “computer B will obey to open terminal and write the message in the command line, and click enter”.
- Why do computer B allow users to give it input to change it’s behavior?
 - Some software allow users to give **input**, this **input** can change the behavior of the software.
 - (imagine, that you tell google “find me this”, and according to your input, it changes it’s behavior to according to your input.)
 - (imagine a theater website, you want to buy tickets in a specific row, you give an **input** to the computer of the theater company with the seat number you want; A malicious input would be to give another input (XCASERASDF) rather a seat number (1..120))
 - Some malicious inputs can cause the computer to run code.
- So... Why do developers want to write software that allow anyone to run code?
 - They don’t. They make mistakes when they write code. They do not think of cases where the **inputs** the user brings, may force the software to behave differently.
 - A malicious message makes a computer do something the owners of the computer didn’t want to be possible to happen. (e.g. accessing another user’s password, someone without permission getting admin privileges, etc)

In next lecture, we will talk about “How can we send a malicious message to a computer, to make it give us the passwords of other users”.

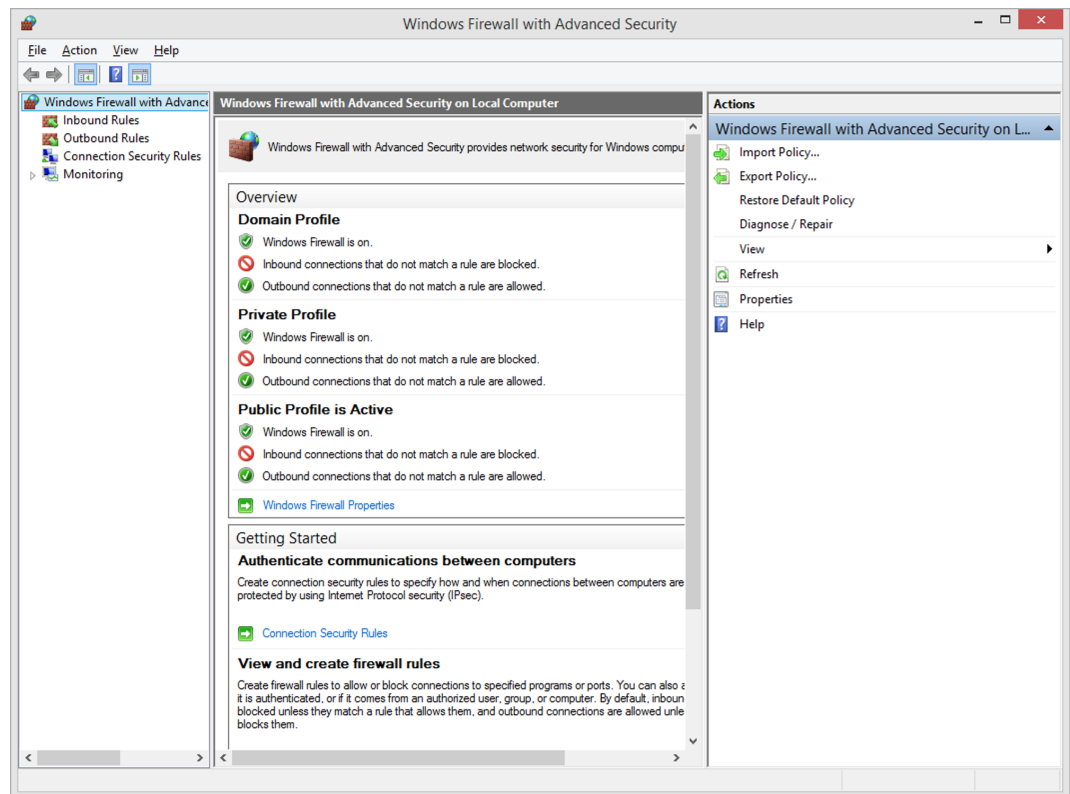
[Learning objective 2: Networks] How does computers communicate with each other?

- We are all connected. There are cables almost anywhere in the world, that connects our computers together.



- What is the address of my computer?
 1. Computers has personal address:
 1. Same as your home address, computers has an address, called IP. Sequence of numbers.
 2. Public IP address: Browse <https://whatismyip.com>
 3. Local IP address: Run command `ipconfig` on windows, or `ip addr` or `ifconfig` on linux/mac.
- Can anyone send my computer a message if they know my IP?

1. Potentially yes, but there are things that blocks it for you.
 1. Firewall can block it.



2. Your router has something called NAT that blocks it under some conditions.

- How a message in the network looks like?
 1. Computer A can send a **Packet** to Computer B, over the network.
 2. Computer B can choose to receive this **Packet**, do something with it, or ignore it.
- Did you know things are talking with your computer right now, without you knowing?
 1. Download Wireshark
 2. Start sniffing the network, and see the **Packets** your computer sends and receives while we are talking.
- What happens when I browse a website? You send a **Packet** to another computer, that answers you with a text message, that your browser transform to a visual website.
 1. Browse <http://vbsca.ca/login/login.asp>
 2. Click "view-source": This is the text the computer responded to you. Your browser takes this text, and makes it a visual interface.
- When I login to a website, what **Packets** my computer send?
 1. Browse <http://vbsca.ca/login/login.asp>

2. Try to submit "username & password".
3. Your computer sent a ****Packet**** to computer B, with the username and password.
4. Computer B chooses to read this **Packet**, wrote a software that extracts the string of "username" and "password" from that textual message, and then it checks if that username and password is correct, and responds with a message to your computer accordingly.
5. Does anyone has an idea why the text message is structured in that way?
 1. There is a **Protocol** called HTTP, that defines "how to chat with a computer that it's purpose is to serve a website (the text we saw in the "view-source")".
 2. Computer B runs a software that parses the **Packet** according to the **HTTP Protocol**, extracts the "username" and "password" fields, and chooses to allow someone to login, or deny.

[Learning objective 3: Servers] How to send a message to another computer? How can I write a software that receives or sends messages (a server)?

- A software that can receive or send messages over the network called server.
- ★ [TRY IT YOURSELF] Write your own **server** in python:
<https://github.com/nattycoder/TCP-CLIENT-SERVER>
[TCP-CLIENT-SERVER-main.zip](#)
- Exercise: Ask what is your friend's Local IP, and send them a message.

HOMEWORK:

- {TODO: WRITE THIS ON THE BOARD} <https://t.ly/kHJe9>
- Please save and open this link (<https://t.ly/kHJe9>), this is your homework, must do before lecture 2, to be able to participate in the workshop, where you will learn a hacking technique and try it by yourself.

[Homework: ULTA cyber lecture 1:](#)

[Learning objective 4: Workshop] How to send a message to a website, and get the user and password details of the admin? (sample website)

- Login to: <https://tryhackme.com/>
- Browse: <https://tryhackme.com/room/sqlinjectionlm>
- If you have done the homework, and understood everything, start from Task 5
- If you haven't done the homework, or need some clarifications, start again from Task 1.
- Let me know if you have any question.

[Bonus] How to learn more? You can learn more by yourself.

- Learn more about cyber by solving cyber puzzles/challenges
 - <https://tryhackme.com>
 - Hack the box
- Learn more about SQL injections, the technique we have learned today.
 - Read this <https://brightsec.com/blog/sql-injection-attack/>
 - Solve all tasks in <https://tryhackme.com/room/sqlinjectionlm>
 - Learn more about SQL and SQL injection here:
https://www.w3schools.com/sql/sql_injection.asp
 - Google "SQL injection tutorials" for more information.